

Chapter 3: Hypoxia & Hyperventilation

Syllabus Source: CAE Oxford (pp. 57-82) | Pass Target: >= 90% AviationExam

1. THE FOUR TYPES OF HYPOXIA (CLASSIFICATION & CAUSES)

Hypoxia Type	Aetiology, Mechanism & Aeronautical Causes
1. Hypoxic Hypoxia (Altitude Hypoxia)	Deficiency in alveolar oxygen partial pressure (P_O2). Most common in aviation. Caused by high altitude, cabin depressurization, or airway obstruction.
2. Anemic (Hypemic) Hypoxia	Reduction in oxygen-carrying capacity of the blood. Hemoglobin is deficient or bound by another agent (Carbon Monoxide poisoning, severe blood loss, smoking, anemia). CO has 200-250x affinity for Hb!
3. Stagnant (Circulatory) Hypoxia	Inadequate regional blood circulation despite normal blood oxygenation. Caused by high positive acceleration (+Gz pulling blood to feet), heart failure, arterial disease, or extreme cold.
4. Histotoxic Hypoxia	Inability of cells to use available oxygen due to cellular poisoning of enzyme systems (cytochrome oxidase). Caused by alcohol, cyanide, narcotics, or carbon monoxide.

2. STAGES OF HYPOXIC HYPOXIA & BLOOD SATURATION

Stage Name	Altitude Range	Arterial O2 Saturation	Clinical Symptoms & Operational Hazards
Indifferent Stage	0 to 10,000 ft	98% down to 90%	Mild night vision impairment starting at 4,000-5,000 ft (rod cells lack O2). No other daytime symptoms.
Compensatory Stage	10,000 to 15,000 ft	90% down to 80%	Body compensates: increased heart rate and respiratory ventilation. Subtle euphoria, impaired complex arithmetic.
Disturbance Stage	15,000 to 20,000 ft	80% down to 70%	Headache, fatigue, loss of self-criticism, cognitive confusion, tunnel vision, cyanosis (blue skin/lips). Loss of fine motor coordination.
Critical Stage	Above 20,000 ft	Below 70%	Rapid incapacitation, deterioration of consciousness, convulsions, coma, followed by irreversible brain death.

3. TIME OF USEFUL CONSCIOUSNESS (TUC) MASTER MATRIX

TUC (also termed Effective Performance Time - EPT) is the maximum time a pilot has from the onset of oxygen deficiency until the ability to perform deliberate, coordinated flight actions is lost:

Flight Level / Altitude	Time of Useful Consciousness (TUC)	Operational State & Pilot Capability
FL 180 (18,000 ft)	20 to 30 minutes	Moderate cognitive decline, euphoria.
FL 220 (22,000 ft)	5 to 10 minutes	Motor impairment, blurred vision.
FL 250 (25,000 ft)	3 to 5 minutes	Rapid loss of reasoning ability.

FL 300 (30,000 ft)	1 to 2 minutes (60-120 s)	Severe confusion, motor collapse.
FL 350 (35,000 ft)	30 to 60 seconds	Extremely rapid incapacitation.
FL 400 (40,000 ft)	15 to 20 seconds	Just enough time to don oxygen mask.
FL 450 and above	9 to 12 seconds	Circulation time from lungs to brain (~10 s).

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Time of Useful Consciousness (TUC) Under Rapid Decompression

SCENARIO (Crucial Flight Physiology Exam Drill): A transport jet is cruising at FL 350 (35,000 ft) when a cargo door seal fails catastrophically, producing an immediate explosive cabin depressurization: - Under normal conditions at FL 350, baseline TUC is 30 to 60 seconds QUESTION: What is the pilot's revised TUC following this explosive decompression, and what mandatory immediate action must be taken? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the Physiological Effect of Explosive Decompression: - Sudden pressure drop forces air out of lungs and creates reverse diffusion (oxygen leaves blood back into lungs!). - Regulatory/Medical Rule: TUC is cut by 50% (half of normal baseline!). Step 2: Calculate the Reduced TUC Window: - Normal TUC at FL 350 = 30 to 60 seconds - Reduced TUC = 50% x (30 to 60 s) = 15 to 30 SECONDS! Step 3: Determine Immediate Pilot Survival Protocol: - 15-30 seconds leaves ZERO time for troubleshooting or communications! - Immediate memory item: DON OXYGEN MASK FIRST (within 5 seconds), establish 100% positive pressure oxygen, then initiate emergency descent to FL 100 / MEA! FINAL ANSWER: Revised TUC is 15 to 30 seconds. The flight crew must don quick-donning masks immediately within 5 seconds before cognitive paralysis occurs!

5. HYPERVENTILATION VS HYPOXIA: DIFFERENTIAL DIAGNOSIS

Hyperventilation is breathing in excess of metabolic needs, causing excessive elimination of carbon dioxide (hypocapnia). This drives blood pH up (Respiratory Alkalosis), causing cerebral vasoconstriction (reduced brain blood supply).

Diagnostic Parameter	Hypoxic Hypoxia	Hyperventilation
Causes	High altitude, depressurization, mask failure.	Stress, panic, anxiety, vibration, motion sickness.
Blood Gas Shift	Severely decreased arterial P _{O2} .	Severely decreased arterial P _{CO2} ; blood alkalosis.
Early Symptoms	Euphoria, false security, loss of judgment.	Anxiety, lightheadedness, cold clammy sweat.
Specific Sign	Cyanosis (bluish skin, lips, nailbeds).	Paresthesia (tingling lips/fingers), Carpopedal spasms (tetany).
Emergency Action	100% O ₂ on, descend below 10,000 ft.	Slow down breathing rate (talk, breathe in paper bag).

[*] OFFICIAL DEFINITION - The Universal In-Flight Emergency Rule

Because the subjective symptoms of hypoxia and hyperventilation overlap (dizziness, tingling, confusion): IF ABOVE 10,000 FT, ALWAYS ASSUME HYPOXIA FIRST! 1. Immediately don oxygen mask (100% O₂ under positive pressure); 2. Check oxygen supply and regulator; 3. Consciously control and slow down respiratory rate to 10-12 breaths/min; 4. Initiate emergency descent if unpressurized.

5. FLIGHT DECK & PASSENGER OXYGEN SYSTEMS

Oxygen Delivery System	Operational User & Altitude Range	Delivery Mechanism & In-Flight Requirements
Continuous Flow	Passengers (Cabin < FL 250)	Fixed rate or barometrically controlled. Rebreather bag. Inefficient for flight crew.
Diluter Demand	Flight Crew (Up to FL 340)	Mixes ambient air with pure O ₂ proportionally with altitude. Delivers 100% O ₂ automatically above FL 320 or when selected.
Pressure Demand	Flight Crew (Above FL 340 / FL 390)	Forces 100% pure oxygen into lungs under POSITIVE PRESSURE to overcome low barometric pressure. Requires active, conscious exhalation!
Quick-Donning Mask	Mandatory for Flight Crew	Must be capable of being placed on face with ONE HAND in LESS THAN 5 SECONDS from the stowed position, accommodating eyeglasses.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the approximate Time of Useful Consciousness (TUC) for a pilot at FL 300 following a gradual decompression?

[A] 10 to 15 minutes [B] 1 to 2 minutes [C] 15 to 20 seconds [D] 9 to 12 seconds

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. At FL 300, normal TUC is 1 to 2 minutes (reduced to 30-60 seconds in an explosive decompression).

Q2: In the event of an explosive decompression at FL 350, how is the pilot's Time of Useful Consciousness affected?

[A] It is doubled due to adrenaline rush [B] It is unchanged [C] It is reduced by one third to one half (30% to 50%) [D] It is extended to 10 minutes

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Explosive decompression reduces TUC by 33% to 50% due to rapid lung exhalation and reverse oxygen diffusion from blood to alveoli.

Q3: Carbon monoxide poisoning causes which specific type of hypoxia?

[A] Hypoxic hypoxia [B] Anemic (hypemic) hypoxia [C] Stagnant hypoxia [D] Histotoxic hypoxia

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Carbon monoxide binds to hemoglobin with ~250 times greater affinity than oxygen, depriving blood of oxygen transport capacity (anemic hypoxia).

Q4: What is the primary physiological consequence of hyperventilation?

[A] Increased arterial P_{CO2} and severe acidosis [B] Excessive loss of CO₂ (hypocapnia) resulting in respiratory alkalosis and cerebral vasoconstriction [C] Immediate lung tissue collapse [D] High arterial oxygen toxicity

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Hyperventilation blows off CO₂, raising blood pH (alkalosis) and constricting cerebral blood vessels, leading to dizziness and tingling.

Q5: If a pilot at FL 250 suddenly feels lightheaded, dizzy, and notices tingling in the fingers, what immediate action should be taken?

[A] Assume hyperventilation, remove oxygen mask, and breathe into a bag [B] Assume hypoxia immediately, don oxygen mask with 100% oxygen, and check oxygen connections [C] Increase cockpit temperature [D] Drink water to restore blood circulation

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Above 10,000 ft, always assume hypoxia first. Don oxygen mask with 100% O₂ immediately before diagnosing hyperventilation.

Q6: At what altitude does night vision acuity begin to deteriorate noticeably due to mild rod cell hypoxia?

[A] At sea level [B] Around 4,000 to 5,000 ft [C] Above 10,000 ft only [D] Above 18,000 ft

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The retina has the highest metabolic oxygen demand in the human body; rod photoreceptors show degraded night vision above 4,000-5,000 ft.

Q7: Why is a pressure-demand oxygen mask required for flight crew operations at altitudes above FL 390?

[A] To prevent oxygen cylinder freezing [B] Because ambient barometric pressure is so low that 100% oxygen at ambient pressure cannot provide sufficient alveolar P_{O2} [C] To allow pilots to speak without a microphone [D] Because continuous flow systems are illegal in Europe

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Above FL 390, total ambient pressure (148 mmHg) is barely above body water vapor and CO₂ pressure (47 + 40 = 87 mmHg); positive pressure breathing is required to force oxygen into blood.